

Consumer Electronics

Ever-larger TV screens with pin-sharp images in brilliant colour, 3D viewing with and without glasses, lightweight yet fast ultrabooks, IT and telecom devices with network capabilities, for the home and on the move—IFA 2012 trade fair (to be held in Berlin, Germany, from 31st August to 5th September) presents trends from every part of the consumer electronics industry. Take a look

1 Ultra high-resolution TVs with brilliant colours

High-definition TV sets have become a part of almost every household and this year's hot topic is 4k. Resolution is now up to 4096×2160 (4k) pixels—equivalent to more than 8 megapixels—so viewers are bound to be amazed by pin-sharp images. With four times the resolution of full HD now available, this technology is ideal for beamers and large screens.

4k panels also make it possible to view stereoscopic films. Compared to conventional full HD screens, these have



OLED TV by Samsung

twice the number of lines. This makes it possible to build 3D TVs that produce images in full HD even when viewed with passive glasses. Even if there are no 4k TV broadcasts on the horizon, this new technology can impressively contribute to further improving image quality, particularly that of large-scale screens.

Another emerging trend in flat-screen technology is the use of organic light-emitting diodes (OLEDs). OLED displays require no additional light source and, compared to conventional LCDs, produce high-contrast images. While LCD cells only work as colour filters, OLED pixels emit light in colour and consequently very impressive images.

OLED displays are very thin, making them ideal for devices with very slim designs. To date they have found their way into mobile phones, smartphones and tablet PCs. Small OLED TVs have been on the market for some time now and later this year the first large (140cm) OLED sets will go on sale.

LG and Samsung have already demonstrated their 140cm (55-inch) OLED TV. While Samsung was reported to launch its OLED-TV at IFA 2012 in Berlin this September, LG was expected to launch its OLED-TV in May itself. TPV Technology, which has taken over the Philips TV brand, will also showcase an OLED TV at IFA 2012.

2 3D with and without glasses

3D technology continues to develop. Glasses for enjoying 3D viewing are now being made with lighter and more



55ZL2 glasses-free 3D TV by Toshiba

stylish frames which can be comfortably worn with normal spectacles.

According to reports, Sony is planning to bring passive 3D TVs to the market. It might even exhibit the first passive 3D displays at IFA 2012. Passive 3D doesn't use any electronics in the glasses. So the glasses are much simpler, lighter and affordable than active 3D glasses.

Also, TV sets are now hitting the market which require no glasses for 3D viewing. Toshiba's ZL2 TV set, for instance, allows 3D viewing without glasses. The TV uses the cutting-edge 4k resolution.

3 Ease of operation

For many users, operating a smart TV still remains a complex affair. For this reason, the industry is urgently working to solve the problem with hardware and software